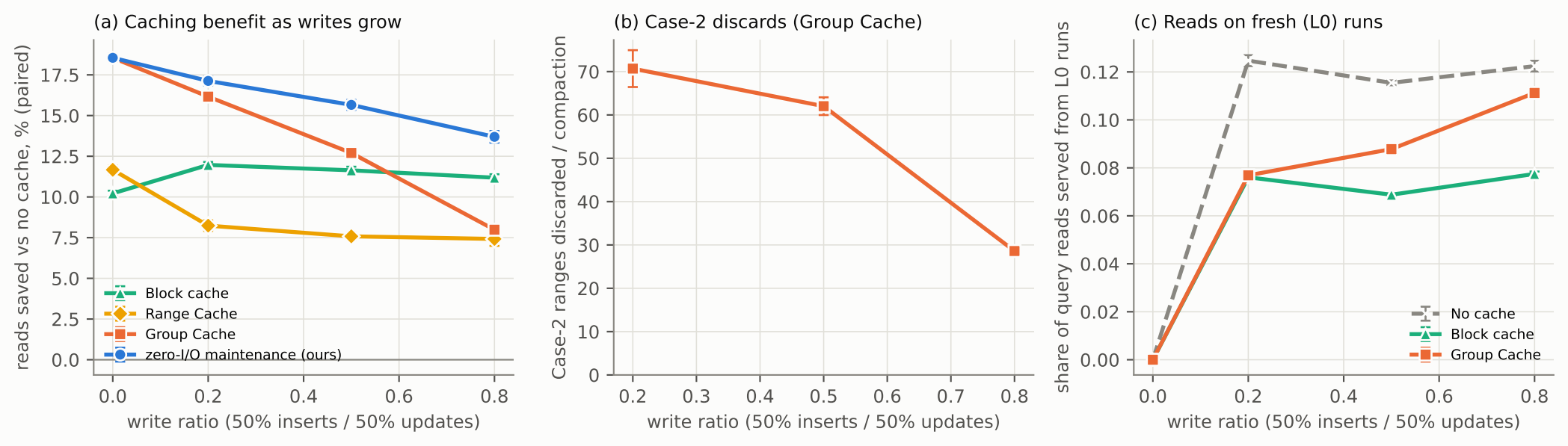
(a) Caching benefit as writes grow
reads saved vs no cache, % (paired)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Block cache
Range Cache
Group Cache
zero-I/O maintenance (ours)
0.0
0.2
0.4
0.6
0.8
write ratio (50% inserts / 50% updates)
(b) Case-2 discards (Group Cache)
Case-2 ranges discarded / compaction
70
60
50
40
30
20
10
0
0.2
0.3
0.4
0.5
0.6
0.7
0.8
write ratio (50% inserts / 50% updates)
(c) Reads on fresh (L0) runs
share of query reads served from L0 runs
0.12
0.10
0.08
0.06
0.04
0.02
0.00
No cache
Block cache
Group Cache
0.0
0.2
0.4
0.6
0.8
write ratio (50% inserts / 50% updates)